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Advanced bearing fault detection at varying rotational speeds using PSO-optimized SVM and CDET feature selection

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Abstract

In this exploration, we introduce a novel intelligent hybrid methodology for multiclass bearing fault detection under varying rotational speeds. Unlike previous studies, our approach combines empirical mode decomposition (EMD), wavelet packet decomposition (WPD), and a comprehensive attribute retrieval strategy from both time and frequency domains. Critically, we present an enhanced distance compensation evaluation technique (CDET) for feature selection, in which the threshold parameter is automatically optimized using particle swarm optimization (PSO) rather than being set arbitrarily. Furthermore, the parameters of both CDET and support vector machine (SVM) classifiers are jointly optimized by PSO, resulting in enhanced classification accuracy and computational efficiency. Extensive experiments on benchmark databases demonstrate that the recommended tactic surpasses conventional tactics regarding fault detection accuracy, stability, and feature compactness. From the original signals, 20 statistical features in the time domain and 6 in the frequency domain were extracted, along with 4 decomposition levels from WPD and the first 6 components from EMD. The outcomes demonstrate that specific features can effectively differentiate various bearing conditions at different speeds, highlighting the efficacy of the recommended approach compared to other fault detection methods.

Keywords: Vibration analysis, Feature selection, SVM, PSO, DPD, EMD

Introduction

Bearings are critical components of rotating machines, and their failure can lead to the complete malfunction of the machinery. Therefore, detecting defects and understanding their characteristics is crucial for effective machine maintenance [1–3]. Various methods have been recommended by researchers to monitor bearing conditions, with vibration signal analysis being particularly significant [4–6]. However, dynamic signals from rotating structures are often nonlinear and nonstationary, rendering traditional time and frequency domain tactics insufficient for detecting various fault conditions. Recently, innovative intelligent approaches in the time–frequency domain have been developed for feature extraction from signals. These tactics typically involve a combination of feature extraction, pre-processing, post-processing, and pattern recognition methods [7–9].

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Among signal analysis methods, the empirical mode decomposition (EMD) method and its enhancements have been extensively utilized in fault diagnosis [10–12]. Wang et al. [13] enhanced the Hilbert-Huang transform by the generalized EMD method, presenting a technique drawing on the correlation between the pulsation signal and its decomposed components to select the most appropriate component for diagnosing rotating machines. Jain et al. [14] introduced a method for detecting gear teeth defects using wavelet denoising and empirical mode analysis. In other studies [15, 16], researchers refined the method of extracting cover curves and improved the empirical mode analysis technique, subsequently applying the resulting algorithms to troubleshoot gears and bearings. Tomar et al. [17] recommended a robust troubleshooting technique for rolling bearings, involving a novel approach that combines denoising with the EMD method, a Bayesian classifier, and thresholding of noisy components. They further utilized wavelet packet transform, Hilbert transform, and signal envelope calculation to develop a technique for identifying bearing conditions. In intelligent fault detection methods, tactics for extracting and selecting suitable features and identifying the system state are used after denoising and signal analysis. Tiwari et al. [18–20] have presented a multi-class classification method to identify defects in gears and bearings through a series of studies. They optimized the parameters of the support vector machine (SVM) by genetic and bee optimization algorithms prior to training and final testing. Statistical meanings including kurtosis, standard deviation, and skewness, extracted from frequency and time domains, were used as input features for the SVM. Tabrizi et al. [21] employed wavelet packet decomposition (WPD) to denoise vibration signals for the early detection of bearing faults. Subsequently, meaningful components of the signal were extracted using ensemble empirical mode decomposition (EEMD). They constructed a feature matrix using the normalized energy feature of the signal components, which was then used as input for the SVM. In other words, to enhance the accuracy of fault detection, researchers have utilized numerous parameters including changed stages of wavelet decomposition, and frequency domain and time domain features related to components obtained via EEMD and EMD approaches. However, with an increase in the count of attributes, the raw feature vector often includes valuable features and irrelevant and redundant ones. Thus, employing a feature selection method becomes essential. In some studies [22–24], the distance compensation evaluation technique (CDET) has been deployed to select the most suitable features from the feature space. Li et al. [25] investigated the classification of bearing defects at six different speeds using SVM. They first extracted 12 time-domain features from the signal and then applied the distance evaluation method to select the most suitable features for input into a multi-class SVM. Dong et al. [26] deployed the component strategy to diminish the feature space after extracting the energy entropy features from the components obtained by the EMD method. They then used the first three principal components as input to an improved SVM with a Morlet wavelet function and particle swarm optimization (PSO) algorithm. Yang and Wu [27] employed a combined approach of general empirical mode analysis, component analysis, and neural networks to detect faults in gearboxes. Ziyani et al. [28] presented a method for selecting the most appropriate set of attributes extracted from the dynamic signals of bearings using Fisher's criterion and the binary PSO algorithm. They then used these selected features as inputs for the SVM. Cao et al. [29] introduced

a two-step technique for bearing fault identification. In the first step, the ReliefF tactic was deployed to select an initial set of all features. In the second step, the best feature set and SVM algorithm constraints were determined to minimize both the bearing condition estimate error and the features number. Sainte et al. [30] suggested a process capable of directly selecting suitable features from the unique feature set while simultaneously optimizing the SVM parameters.

In most feature selection algorithms, including CDET, there is a bound called the threshold that theatres a crucial role in determining the optimal features. It is important to note that the value of this parameter has often been determined by trial and error in several studies. In this exploration, after analyzing the signal using wavelet packet transformation and empirical mode analysis, frequency and time domain characteristics associated with the components and levels obtained from the analysis were extracted to form the feature matrix. Owing to the high dimensionality of the resulting feature matrix, the CDET algorithm was used to choose the suitable features.

Although a variety of intelligent diagnostic tactics have been developed for bearing fault detection, many existing methods suffer from high computational cost and limited accuracy owing to the presence of redundant and irrelevant features in high-dimensional feature spaces. Traditional feature selection approaches, including CDET, rely on manually determined threshold values, which can compromise the effectiveness of the selected features.

In this research, we address these limitations by introducing an automated and optimized feature selection framework. Specifically, the critical threshold parameter in CDET is automatically determined through PSO, ensuring the selection of the most discriminative features for fault classification. Additionally, by jointly optimizing SVM classifier parameters with the same PSO process, our approach significantly improves both detection accuracy and computational efficiency. To the best of our knowledge, this is the first work that systematically integrates PSO-based optimization for both feature selection and classifier parameter tuning in the context of multi-class bearing fault detection at variable rotational speeds. Our comprehensive experiments confirm the superiority of the recommended tactic over established baselines. In this exploration, the optimal value of the threshold parameter in CDET and the best parameters of SVM were obtained by PSO to minimize both the detection error of the bearing conditions and the dimensions of the feature vectors. The outcomes demonstrate that the recommended technique is highly effective in identifying the type and severity of defects in the bearings. The remaining sections of this article are outlined as follows: the second section describes the laboratory setup. The third section explains the troubleshooting methods utilized in this exploration. The fourth section provides detailed information on the recommended tactic. The fifth section analyzes the outcomes of the work. Finally, the article is concluded in the sixth section.

Laboratory setup

For the dynamic signals in this work, the laboratory database obtained from the Case Western Reserve University (CWRU) website [31] is used. These data have been widely utilized by researchers to evaluate their recommended algorithms. As shown in Fig. 1,

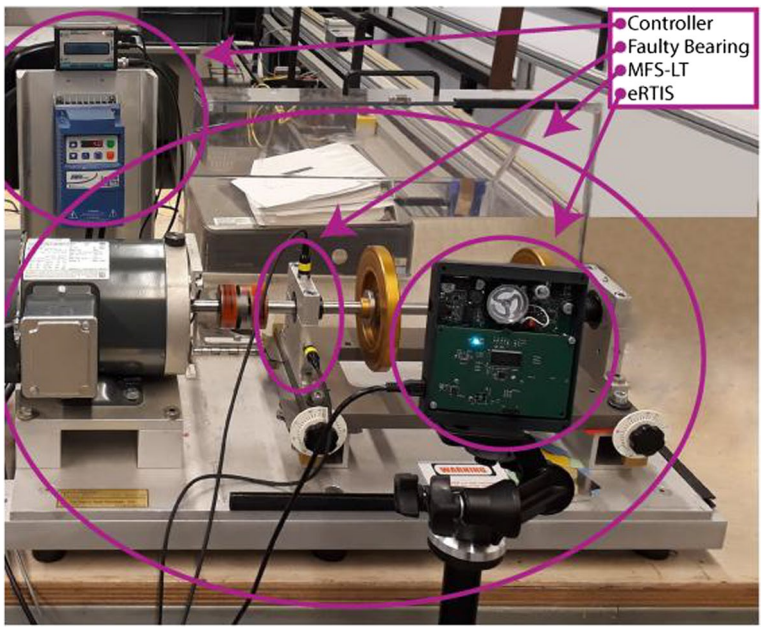


Fig. 1 Laboratory set for detecting defects in bearings [32]

Table 1 Rolling bearing details [32]

Parameter	Value (mm)
Inner rim diameter	25
Outer rim diameter	50
Thickness	14
Bullet diameter	39

this setup contains a motor (left), a dynamo motor (right), and a coupling (middle). The defects in the bearings were induced using electrical machining methods.

The dynamic signals used in this exploration are associated with the SKF6208-4RS JEM bearing. The specifications of this bearing are planned in Table 1. Defects were introduced in the ball elements, inner ring, and outer ring with a depth of 0.279 mm and varying diameters of 0.1778, 0.3556, 0.5334, and 0.7112 mm. Vibration data for four diverse loadings (0, 1, 2, and 3 HP) at rotational speeds of 1798, 1770, 1755, and 1730 rpm were collected using an acceleration sensor with a sampling rate of 12 kHz. This exploration considers four states: healthy state, defective ball, faulty inner ring, and faulty outer ring to investigate the multi-class defect state.

Methods

In this section, the methods employed in this exploration will be briefly described.

EMD

EMD is designed for examining nonstationary and nonlinear signals. The signal is iteratively decomposed into Intrinsic Mode Functions (IMF)s through a process known as sifting. During sifting, local maxima and minima are identified, and a mean envelope

is created by interpolating between these points. The mean envelope is deducted from the original signal, resulting in the first IMF. This process is repeated on the residual signal to derive subsequent IMFs until the final residual is a monotonic function with one extremum. This method adaptively decomposes any complex signal into a series of IMFs, ranging from high-frequency to low-frequency components. The decomposed signal can be expressed as [33]:

$$x(t) = \sum_{i=1}^{n} c_i(t) + r_n(t) \tag{1}$$

which $c_i(t)$ shows the ith function of the intrinsic mode and $r_n(t)$ is the residual of the signal $x(t)$. Each IMF contains the following conditions:

1. In the entire database, the count of extreme points (maximum and minimum) and the count of zero crossing points should be equal or at most one different.

2. At each point, the average of the fitted patch on the local maximum points and the fitted patch on the local minimum points must be zero.

Feature extraction

Owing to the nonlinear nature of bearing dynamics and the nonstationary nature of vibration signals, examining only the time series related to the vibration data does not provide sufficient info about the size and type of the defect. Therefore, different feature removal approaches are utilized in this exploration. To form the feature vector for each vibration signal, the following steps are taken:

1. Time-domain statistical features: - From each raw signal, statistical features in the time domain are extracted according to Table 2. These features are labeled F1–F20.
2. Empirical mode decomposition (EMD) features:- All pulsation signal is disintegrated into different components by the adaptive EMD procedure. Since the first little components (IMFs) contain extra info about the defect, the first five components are selected in this exploration.
   - For each selected IMF, the statistical features from Table 2 are extracted, resulting in features labeled F21–F120.
3. Frequency-domain features:Frequency-domain features: - From each raw signal, frequency domain features are obtained according to Table 2, labeled F121–F124.

Table 2 Some of the characteristics in the field of time and frequency

$F_1 = \frac{1}{N}\sum_{i=1}^{N} x(i)$	$F_2 = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x^2(i)}$	$F_3 = \frac{x_{RMS}}{N^{-1}\sum_{i=1}^{N} \lvert x(i)\rvert}$	$F_4 = \max(x)$
$F_5 = \max(\lvert x\rvert)$	$F_6 = \frac{x_{Peak}}{N^{-1}\sum_{i=1}^{N} \lvert x(i)\rvert}$	$F_7 = \min(x)$	$F_8 = \frac{x_{Peak}}{\left(N^{-1}\sum_{i=1}^{N} \lvert x(i)\rvert\right)^2}$
$F_9 = x\left(\frac{N+1}{2}\right)$	$F_{10} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} (x(i)-\overline{x})^2}$	$F_{11} = \log\left(F_9 + \frac{\mathrm{RMS}}{0.078}\right)$	$F_{12} = \int \lvert x\rvert^2 dt$

- The same frequency domain features are calculated for the selected IMFs, resulting in features labeled F125–F144.

4. vWavelet packet decomposition (WPD) features:- Wavelet packet analysis is employed as a feature extraction tool, effective in analyzing both low and high-frequency components of nonstationary signals.

   - The choice of the mother wavelet significantly affects the efficiency of the WPD decomposition method. According to research [20], the db44 wavelet is identified as the most suitable for identifying bearing and gearbox defects.

   - In this exploration, the db44 wavelet is deployed to analyze vibration signals. The extreme depth of the decomposition tree is set to 3, as shown in Fig. 2.

   - For each of the fourteen coefficients obtained from the WPD, the statistical characteristics in the time domain (Table 2) are extracted, resulting in features labeled F145–F424.

   By combining these comprehensive feature extraction methods, a detailed feature vector is constructed for each vibration signal, facilitating the effective identification and classification of bearing defects. This comprehensive feature vector provides a detailed representation of the vibration signal, capturing various aspects of the signal's characteristics. By using these diverse feature extraction methods, the recommended approach can better identify and classify the type and severity of defects in the bearings.

   Following the extraction of features from all the signals, the resulting feature matrix is structured as follows:

$$X = [f_{m,c,j}] \tag{2}$$

In which **X** is the feature matrix. M remains the count of vibration signal samples. N displays the overall count of extracted features (in this case, 424). F_{mcj} displays the jth feature extracted from the cth vibration signal sample. This feature matrix is used for further analysis, including feature selection and classification, to identify the bearing situations and detect faults effectively. The feature number corresponding to each of the above feature extraction approaches is presented in Table 3. In this paper, the

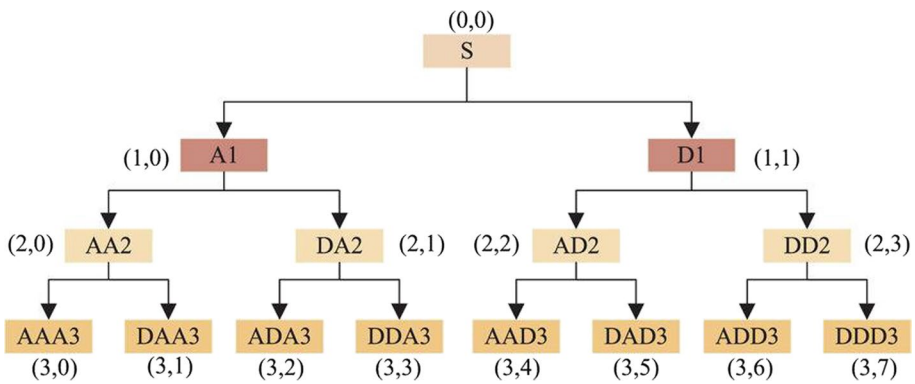


Fig. 2 Decomposition tree of wavelet packet transform at level 3 [34]

Table 3 Number of features for each extraction method

Feature extraction method	Number of features
Time-domain statistical features	20
EMD component features	100 (5 IMFs × 20)
Frequency-domain features	24 (4 raw + 5 IMFs × 4)
WPD features	280 (14 coefficients × 20)
Total	424

total number of features for every signal sample is $J = 424$, and the count of examined conditions, C, corresponds to the different classes of defects or various defect sizes.

Improving the distance compensation evaluation method to select the most suitable features

When the feature space obtained from the training documents is highly dimensional, the feature matrix will contain not only fault-sensitive features but also nonsensitive and redundant ones. Using a high-dimensional feature matrix as input for various classifiers leads to increased computation time and reduced efficiency and accuracy of intelligent methods in predicting the state of a system. To address the problem of reducing the feature space and selecting effective features, recent research has employed various feature selection approaches, including Fisher discriminant analysis, principal component analysis, and the CDET. These methods help to identify and retain the most relevant features, thereby improving the performance of classifiers and reducing computational overhead.

In this exploration, an improved method for the CDET exists and is applied to choose the most suitable features from vibration signal data. Initially, features are extracted using time-domain, frequency-domain, EMD, and WPD approaches to form a high-dimensional feature matrix. Distance compensation scores are then calculated for each feature to evaluate their sensitivity to different fault conditions. An optimal threshold value, crucial for feature selection, is determined and optimized using the PSO algorithm to minimize prediction error and the number of features. Features with scores exceeding this threshold are selected, and these selected features are used to train a classifier, which is validated with test data to confirm the effectiveness of the feature selection procedure. This approach enhances the accuracy and efficiency of fault recognition and classification. The steps of the feature selection process with the CDET method are as follows:

1. The ordinary distance consistent with the j-th feature for signals that have class c information is calculated as follows:

$$d_{c,j} = \frac{1}{(M_c - 1)M_c} \sum_{I,m=1}^{M_c} \left| f_{m,c,j} - f_{I,c,j} \right|, \qquad m, I = 1, 2, 3, ..., M_C \tag{3}$$

Then, the average distance for all classes is calculated from the following equation:

$$d_j^{(w)} = \frac{1}{C}\sum_{c=1}^{C} d_{c,j} \tag{4}$$

2. The variance factor corresponding to the j-th property of $d_j^{(w)}$ is defined and calculated as follows:

$$v_j^{(w)} = \frac{max\left(d_{c,j}\right)}{min\left(d_{c,j}\right)} \tag{5}$$

3. The average j-th characteristic of all signals for class c is calculated from the following equation:

$$u_{c,j} = \frac{1}{M_c}\sum_{m=1}^{M_c} f_{m,c,j} \tag{6}$$

Then, the average distance between signals belonging to diverse classes is calculated as follows:

$$\mathrm{d_j^{(b)}} = \frac{1}{\mathrm{C(C-1)}}\sum_{\mathrm{c,e=1}}^{\mathrm{c}} \left|\mathrm{u_{c,j} - u_{e,j}}\right|, \qquad \mathrm{c, e = 1, 2, ..., C} \tag{7}$$

The variance factor corresponding to the j-th feature of $d_j^{(b)}$ is calculated from the following equation:

$$v_j^{(b)} = \frac{\max\left(\left|u_{e,j} - u_{c,j}\right|\right)}{\min\left(\left|u_{e,j} - u_{c,j}\right|\right)}, \quad c, e = 1, 2, ..., C \tag{8}$$

4. The correction factor for the *j*th feature is defined according to the following relationship:

$$\lambda_j = \frac{1}{\dfrac{v_j^{(w)}}{max\left(v_j^{(w)}\right)} + \dfrac{v_j^{(b)}}{max\left(v_j^{(b)}\right)}} \tag{9}$$

5. The coefficient of $\overline{\alpha}_j$, which is an index to evaluate the characteristics of j-th, is defined below:

$$\overline{\alpha}_j = \frac{\alpha_j}{\max\left(\alpha_j\right)} \tag{10}$$

where in

$$\alpha_j = \lambda_j \frac{d_j^{(b)}}{d_j^{(w)}} \tag{11}$$

6. The index of $\overline{\alpha}_j$ is calculated for all features, and then, it is compared with the threshold value $\xi \in [0, 1]$ that is predetermined. If $\xi < \overline{\alpha}_j$, the j-th feature is selected as an effective and appropriate feature for the troubleshooting process, and otherwise, it is removed from the feature matrix. One of the significant challenges of the CDET method is selecting the appropriate parameter ξ. While a value of 0.5 has been used in a previous study [35], choosing an arbitrary value for ξ can either remove features that contain useful information or select features that are redundant and insensitive to defects. To address this, a method will be presented in Section 4 to determine the optimal value for ξ using the PSO procedure. By calculating the optimal ξ, the most appropriate subset of the feature space is selected. This method, referred to as the Optimized Distance Compensation Evaluation Technique (OCDET) in this paper, ensures more effective feature selection.

SVM

The SVM algorithm studies a conventional database called the training database and aims to generalize this knowledge to make accurate predictions on new data. The basic concepts of the support vector machine method are described below.

Consider data x_i $(i=1,2,..,n)$ categorized as positive $(y_i=+1)$ or negative $(y_i=-1)$ classes. The optimal separator hyper-plane of $f(x) = wx + b$ can be achieved by solving the optimization problem:

$$\min\left(\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{N}\xi_i\right) \quad \text{subject to}: y_i(wx+b) \geq 1-\xi_i \tag{12}$$

which ξ_i measures the space between the hyper-plane and the samples placed on the wrong side of the hyper-plane, and C is the penalty parameter. By introducing the Lagrange coefficients and for the constraints of the problem of Eq. (12) and determining the saddle equations, a quadratic optimization problem is obtained. If there is an optimal value for the parameter α_i^* that is obtained by solving this equation, then the optimal parameters of the hyper-plane can be calculated. Finally, the classification function can be written as follows:

$$f(x) = sign\left(\sum_{i=1}^{N}\alpha_i y_i(x_i x_i) + b\right) \tag{13}$$

In cases where the organization is nonlinear in the input space, the SVM method can be applied by mapping the data to the feature space with higher dimensions. In this scenario, a nonlinear function is deployed to map n-D input vector to l-D feature domain, resulting in the following modification to the decision function:

$$f(x) = sign\left(\sum_{i,j=1}^{N}\alpha_i y_i K\left(x_i, x_j\right) + b\right) \tag{14}$$

in which $K\left(x_i, x_j\right)$ stands for a kernel function and

$$K\left(x_i, x_j\right) = \left(\phi(x_i).\phi(x_j)\right) \tag{15}$$

The kernel function that is generally used in practical applications of SVM is the radial basis function, which is defined as follows:

$$K\left(x.x_j\right) = \exp\left(-\frac{\left\|x - x_j\right\|^2}{\sigma^2}\right) \tag{16}$$

The selection of penalty parameters C and σ strongly influences the performance of SVM. As you will see in this work, the PSO optimization procedure will be applied to discover the best values of parameters.

Sometimes, we encounter problems involving more than two classes, including fault detection in rotating machines with multiple fault sources. In such cases, two strategies are typically used to construct multi-class SVM: "one-to-all (OAA)" and "one-to-one (OAO)." In this article, only the "one-to-one" strategy is employed.

PSO algorithm

The PSO procedure is an optimization procedure inspired by the social behavior of fish and birds. This method involves a population of solutions, known as particles, which navigate the solution space to locate the optimal solution. Every particle updates its location based on its own experience and the experiences of neighboring particles, striving to determine the optimal parameters for a given problem. In this exploration, the PSO procedure is applied to determine the optimal value of the parameter ξ in the CDET technique and the optimal parameters of the SVM classifier. By doing so, the PSO procedure helps minimize the estimated error and the count of selected features, enhancing the accuracy and efficiency of fault detection and classification.

The PSO procedure is a population-based optimization method stimulated by the social performance of groups of fish and birds [36–38]. In this algorithm, entities called particles are used to find the greatest answer in the search domain. Every particle displays a potential solution to the problem. For a d-dimensional problem, each particle has two vectors: a speed vector and a position vector, as follows:

- **Position vector:** Displays the current location of the ith particle in the solution space.
- **Velocity vector:** Denotes the rate of change of the location of the ith particle.

The PSO procedure iteratively updates the location and speed of each particle based on the particle's own experience and the experience of its neighbors. This process guides the particles toward the optimal solution. The update rules for the velocity and position vectors are typically governed by the following equations:

$$\begin{aligned} &Position: \ \vec{X}_i = \{X_{i1}, X_{i2}, ..., X_{id}\} \\ &Velocity: \ \vec{V}_i = \{V_{i1}, V_{i2}, ..., V_{id}\} \end{aligned} \tag{17}$$

In the initialization phase, an initial location and an initial speed are haphazardly assigned to each particle. In the calculation phase, each particle deploys its personal best experience

($\vec{X}_{pBest}$) and the best experience of all particles ($\vec{X}_{gBest}$) to find its next location and move in the search domain as:

$$\vec{V}_i(t+1) = w\vec{V}_i(t) + c_1 r_1\left(\vec{X}_{pBest} - \vec{X}_i(t)\right) + c_2 r_2\left(\vec{X}_{gBest} - \vec{X}_i(t)\right) \tag{18}$$

$$X_I(t+1) = X_i(t) + V_i(t+1) \tag{19}$$

which $\vec{V}_i(t+1)$ is the speed vector of the particle in cycle $t+1$ and the $\vec{X}_i(t+1)$ is the location vector of the ith particle. c_1 is the personal learning coefficient, and c_2 is the social learning coefficient. r_1 and r_2 have random values in [0, 1]. In this work, the values of c_1, c_2, and w are considered according to a reference.

In the PSO algorithm, the location and speed of the particles are created haphazardly. Then, by computing the objective function value, $\vec{X}_{pBest}$ are determined for every particle and $\vec{X}_{gBest}$ are determined for the complete population. In every cycle, the values of parameters c_1, c_2, and w are calculated. The velocity and position of each particle are updated by Eqs. (17) and (18) in PSO. After determining the new position of the particles, the value of the objective function is calculated for each new particle. If the objective function value in the new particle is less than the value of the cost function in that particle, then $\vec{X}_{pBest}$ it will be updated. Finally, the cost function values $\vec{X}_{pBest}$ and $\vec{X}_{gBest}$ are compared and the lowest of them is selected as the overall optimum in the current cycle. This process continues until the termination condition is met.

Suggested intelligent technique

In this exploration, a novel intelligent hybrid technique for multi-class fault identification of bearings is presented. This method combines EMD, feature extraction, wavelet packet transform, and improved SVM-OAO (support vector machine—one against one) and CDET algorithms. In the recommended hybrid algorithm, a wide range of features from the frequency domain, time domain, and time–frequency domain related to raw signals, components resulting from EMD analysis, and different levels of wavelet packet transformation are extracted. The recommended OCDET algorithm is then applied to select the most suitable subset of these features. Finally, these optimal features are deployed as input to an improved SVM optimized with PSO. The flowchart of the technique offered in this article is displayed in Fig. 3. The stages of the suggested technique are as follows:

1. Data splitting:- 60% of the database is given to "training data for parameter estimation," and the remaining 40% is used as "final test data."
2. Signal decomposition:- Each signal from the "training data for parameter estimation" set is decomposed into simpler components using the EMD method.
3. Feature extraction:- For each signal sample, features from the time and frequency domains, and the coefficients obtained from wavelet packet transformation are extracted, as described in Sects."Laboratory setup"and"Methods". This results in 424 features for each signal.

4. Feature matrix formation:- The feature matrix is constructed with rows corresponding to signals and columns corresponding to features. The data in each column is then normalized to have values in the range [0, 1].
5. Database separation:- The normalized training data is split into two databases: "training data" and "validation data." The training database is deployed to build the SVM classifier, whereas the validation database is deployed to train and tune the classifier.
6. The PSO optimization algorithm is deployed to find the best C values and Swarm size (S) parameters in the SVM procedure and threshold parameters in the CDET feature selection algorithm. For this aim, the objective function is defined as follows:

$$\text{Objective} = \left[(\alpha - 1)\frac{\text{Incorrectly classified}}{\text{Classified samples}} + \alpha\frac{R}{N}\right] \times 100 \tag{20}$$

where R is the feature number selected in each optimization step and N stands for the total feature number. According to a study [16], the S parameter is considered equal to 0.01. The higher and lower limits of the decision variables are listed in Table 4.

7. To evaluate the recommended tactic, the final test data has been used. For this purpose, with the help of the optimal classifier obtained in step 6, the data class and, as a result, the conditions governing the bearing are detected.

By following these steps, the recommended tactic ensures that the most appropriate features are selected and used to build a robust classifier for multi-class fault identification of bearings.

Outcomes

Vibrational data

To assess the performance of the suggested technique, the vibration data collected from the laboratory setup depicted in Fig. 1 have been utilized. Detailed explanations related to the study cases used in this article are presented in Table 5.

- Case 1: This case has been employed to measure the competence of the suggested technique in identifying the type of defect with a diameter of 0.1778 mm at different rotational speeds. This scenario allows for the assessment of the method's sensitivity to defect types under varying operational conditions.

- Cases 2 to 4: These cases are designed to identify different sizes of defects at various rotational speeds. Each case considers vibration data corresponding to defects with depths of 0.1778, 0.3556, 0.5334, and 0.7112 mm. This setup helps evaluate the method's ability to distinguish between varying severities of defects.

Table 4 The range of changes in design variables

Parameter	Lower	Upper
C	10^{-4}	10^{4}
σ	10^{-4}	10^{4}
ξ	0	1

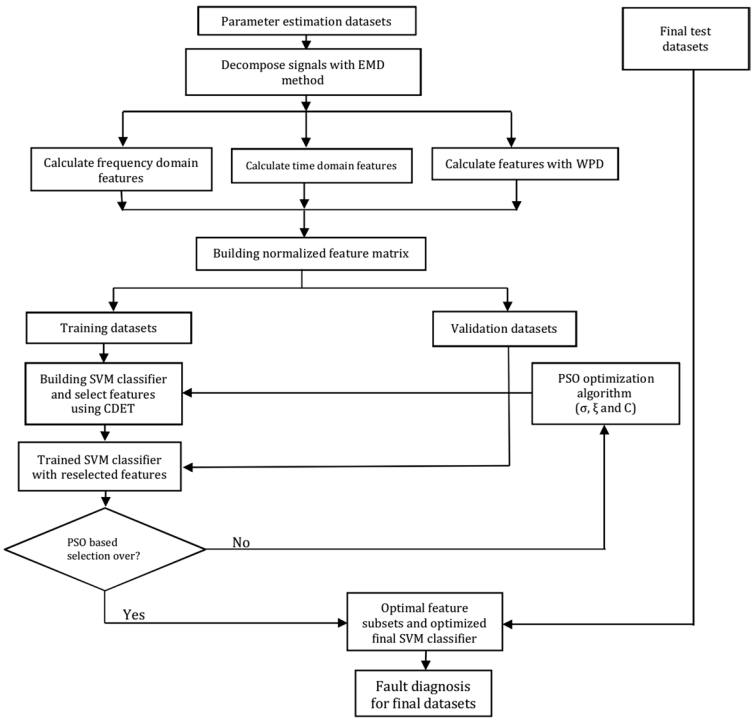


Fig. 3 Flowchart of the suggested troubleshooting method

For example, Fig. 4 shows the signals equivalent to different conditions for two defect sizes of 0.1779 mm and 0.5335 mm at a rotational speed of 1750 rpm. As can be observed in this diagram, the characteristics of shocks and the amplitude of vibration signals differ under various conditions and defect sizes. This variation in signal characteristics illustrates the challenges faced when trying to diagnose the category and severity of defects by merely examining these raw signal features. Given these complexities, it becomes apparent that accurately diagnosing the type and severity of a defect solely based on the raw vibration signals is extremely difficult. The variations in amplitude and shock characteristics can provide some information, but they are insufficient for precise fault detection. Therefore, there is a necessity to extract detailed features from these signals and develop an intelligent fault detection method that can automatically and reliably identify the faults.

The suggested technique addresses this by decomposing the dynamic signals using EMD, extracting comprehensive features from the frequency, time, and time–frequency domains, and applying advanced feature selection and classification tactics. This approach not only enhances the accuracy of fault recognition but also allows for the automatic identification of defect types and their severities, thereby overcoming the limitations of direct signal examination.

Choosing the appropriate feature using the suggested technique

According to the flowchart in Fig. 3, by using the EMD technique, the dynamic data used for training the SVM network and selecting the most proper features are analyzed. For

example, the first five components resulting from the examination of the dynamic signals in Figs. 4c and 4e, which correspond to the inner ring and outer ring defects with a size of 0.1778 mm, are shown in Fig. 5. These components, known as IMFs, are derived from the EMD process and represent different frequency bands of the original signal. The decomposition into IMFs allows for a detailed examination of the signal's characteristics at various scales. Each IMF captures specific aspects of the signal's behavior, which are crucial for identifying and diagnosing faults. By analyzing these components, it becomes possible to extract relevant features that are subtle to the existence and severity of faults. These features are then used to train the SVM network, ensuring that the classifier can accurately distinguish between different kinds of faults and defect sizes. The selection of the most appropriate features is a critical step in improving the enactment of the fault diagnosis system, making it more reliable and effective in real-world applications.

After analyzing the signals, the featured medium is designed, and with the help of the PSO optimization algorithm, the best suitable features and the best parameters of SVM are determined simultaneously to minimize the objective function given by Eq. (19). For example, in Fig. 6, the distance compensation evaluation index α is presented for all features in two case studies: inner and outer ring defects. The feature corresponding to $\alpha = 1$ is considered the most sensitive parameter for defect classification. The optimal threshold values obtained by the PSO algorithm for inner and outer ring defects are 0.6966 and 0.2333, respectively. According to Fig. 3, features 128 F (root mean square frequency of the first IMF) and 129 F (frequency mean of the second IMF) are identified as the most sensitive feature vectors for the two study cases. The distribution of these two features for different sizes of inner and outer ring defects is shown in Figs. 6a and b, respectively. It is evident from these figures that selecting the feature vector using the recommended tactic allows for a clear distinction between different bearing conditions. Additionally, Fig. 7 shows that the distance between the healthy state and various defect sizes in the inner and outer rings is significant. This separation aids in easily distinguishing between healthy and defective classes, enhancing the accuracy of the fault diagnosis system.

Comparison with other fault detection methods

To estimate the effectiveness of the suggested technique, it is essential to compare its performance with other existing troubleshooting methods. Various tactics have been developed and employed for fault identification in bearings, each with its strengths and limitations. The following comparisons highlight the advantages and improvements offered by the suggested hybrid process.

To compare the capabilities of the feature choice and fault recognition technique suggested in this article with other methods, three different approaches have been used, as described in Table 6. In the first method [39], three numerical features—standard deviation, elongation, and skewness—are extracted from the raw signals. The SVM classification method is then optimized using the genetic algorithm (GA). For this optimization process, the population size and the maximum number of cycles are both set to 50. In the second method [21], the signals are de-noised using

the WPD technique. The normalized energy from the first three IMFs obtained by the EEMD method is used as the input feature vector for the SVM.

In the third method [40], optimal features from the n-D original signal feature matrix and the best parameters of SVM are selected via the binary PSO algorithm. The count of cycles for the BPSO algorithm is set to 50, and the count of particles in each cycle is set to 20.

Since optimization algorithms have a random nature and the obtained outcomes may vary with each execution, each of the methods (one to three) and the recommended tactic in this exploration were implemented 30 times for troubleshooting the study cases described in Table 5. Finally, the outcomes connected to the correctness of fault recognition in the training and testing stages are reported in statistical terms as "mean accuracy of fault prediction (mean)" and "standard deviation of fault prediction outcomes (std)" and are presented as mean $\pm$ std in Table 7. Additionally, the optimal parameters associated with the SVM procedure, the threshold parameter ξ, and the optimal features are also presented in Table 7. By presenting these comparative outcomes, the effectiveness of the suggested technique in feature selection and fault recognition can be evaluated against other established approaches.

As seen in Table 7, the highest error in identifying fault in all cases (1 to 4), both in the training and testing, is related to the second method. Additionally, the outcomes obtained from the second method exhibit the highest standard deviation, indicating greater dispersion of the answers compared to other methods. This method uses the energy feature vector related to the components obtained from the EEMD method to form the feature matrix, which results in significant error. In contrast, the first method, which utilizes elongation, skewness, and standard deviation, shows a significantly reduced defect identification error. This can be attributed to the sensitivity of these features to the presence of faults in various components of the bearing. The training error in the first method is zero in all cases except for the third case, troubleshooting. Furthermore, in the second case study, the correctness of the first technique in predicting the final test data is higher than all other methods, and it is very close to the method recommended in this article. Overall, the recommended tactic demonstrates superior performance by combining effective feature extraction tactics and optimization algorithms, leading to more accurate and reliable fault detection compared to the other approaches evaluated.

Table 5 Explanation of the data considered in the fault detection process

Case	Class number	Test number	Training number	Working state	Fault size (mm)
Fault type	4	48	32	Intact	-
Detection of different sizes of inner rim defects	5	60	40	Inner rim	0.1778–0.7112
Detection of different sizes of outer rim defects	4	48	32	Outer rim	0.1778–0.5334
Detection of different sizes of rolling element defects	5	60	40	Rolling element	0.1778–0.7112

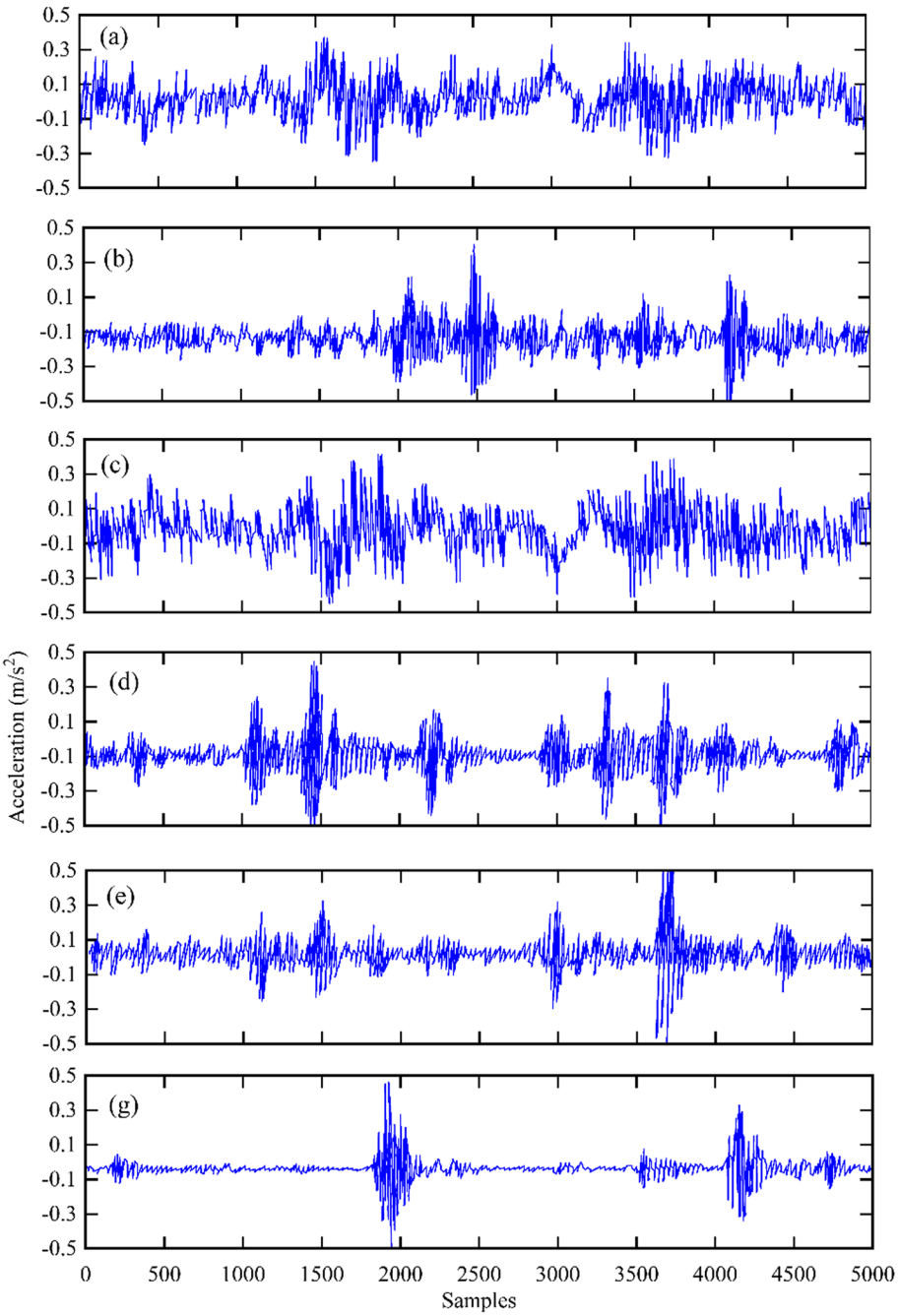


Fig. 4 Vibration signals for four different modes: **a** intact, **b**, **c** inner rim with depths of 0.1778 mm, **d** inner rim with depths of 0.5334 mm, **e** outer rim with depths of 0.1778 mm, and **g** ball element with depths of 0.1778 mm

According to Table 7, the suggested method has demonstrated high accuracy in predicting the data for the first, second, and third study cases during both the validation and training phases across all 30 implementations. The outcomes clearly show the superiority of the suggested technique in predicting the final test data for the first, third, and fourth study cases. Additionally, this method shows the lowest standard deviation in data

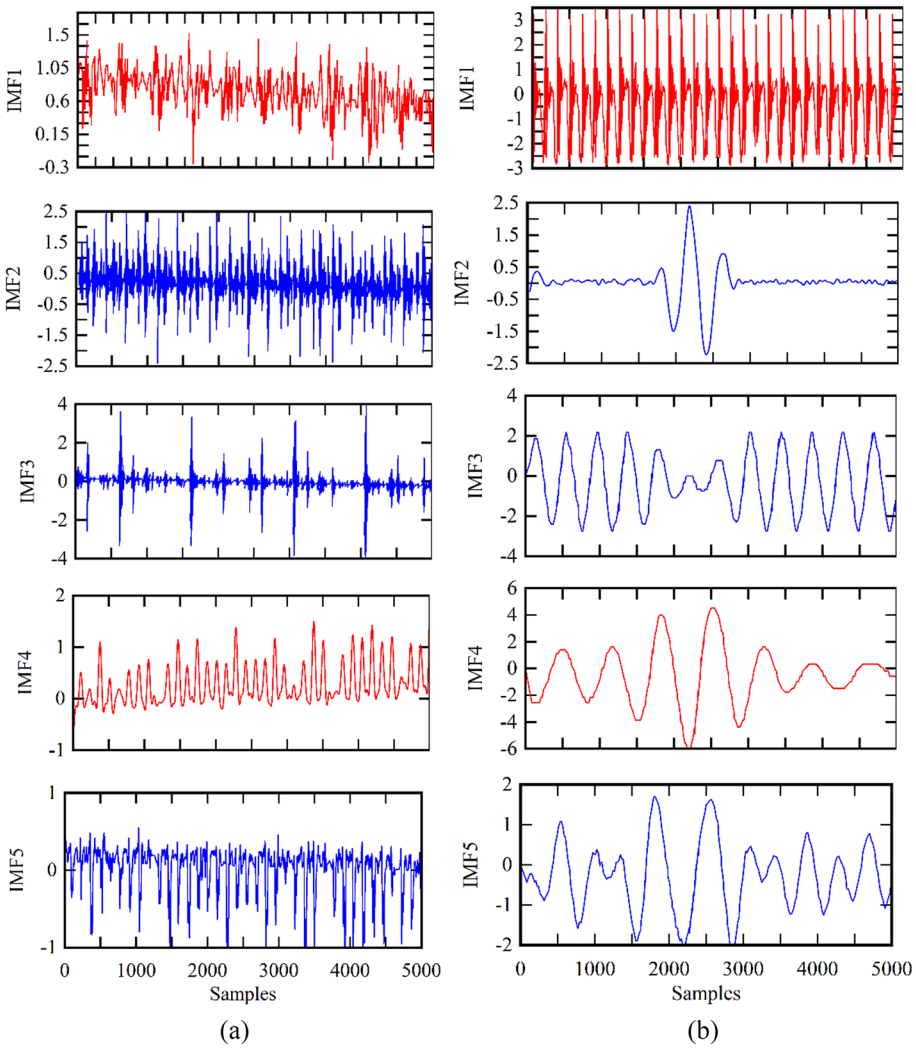


Fig. 5 The components obtained from the analysis of the vibration signal by the EMD method: **a** inner rim defect with a depth of 0.1778 mm and **b** outer rim defect with a depth of 0.1778 mm

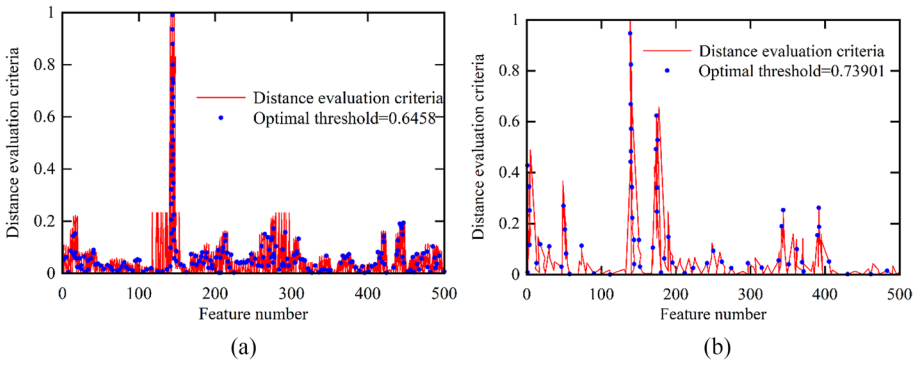


Fig. 6 Distance evaluation criteria for all features and optimal threshold value: **a** internal rim defect and **b** external rim defect

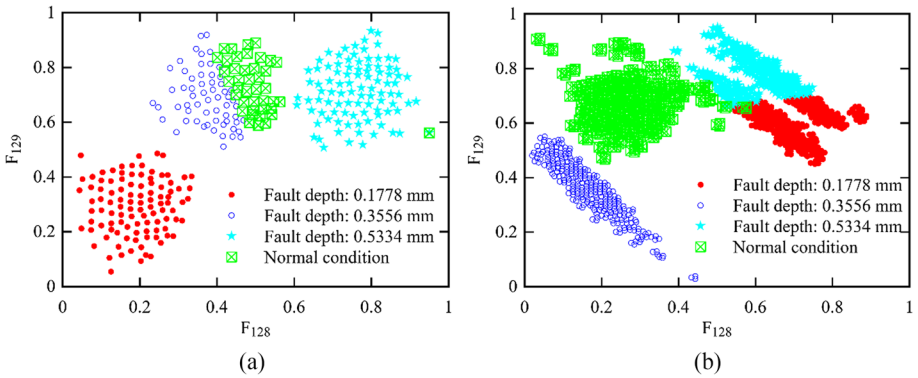


Fig. 7 Distribution of optimal features obtained by the suggested technique: **a** internal rim defect and **b** external rim defect

Table 6 Description of comparative methods

Method	Description
Ref [39]	The features of elongation, skewness, and standard deviation have been used to extract the feature. Then, with the help of the support vector machine algorithm, the state of the system is classified. A genetic algorithm has been used to determine the classifier parameters
Ref [21]	Signals are decomposed into different components by means of noiseless wavelet packet transformation and a general experimental mode decomposition algorithm. Each feature vector contains the energy characteristic of the first three natural modes of each signal, which is normalized. Then, the support vector machine algorithm has been used to detect the system state. The parameters of the support vector machine are determined with the help of the cross-validation algorithm
Ref [40]	The optimal feature set is directly derived from the original feature set using binary PSO, determining and simultaneously optimizing the support vector machine parameters for fault detection

Table 7 Outcomes of fault detection accuracy and optimal parameters

Case	Method	Optimization parameter	Number of features	Accuracy in training	Accuracy in testing
Case I	Ref. [39]	$C=0.2133$ $\sigma=0.6519$	–	99.98	95.65
	Ref. [21]	$C=0.7787$ $\sigma=1.2350$	–	84.62	82.05
	Ref. [40]	$C=5.0052$ $\sigma=2.0382$	182	99.75	96.78
	Recommended tactic	$C=15.3628$ $\sigma=0.8501$ $\xi=1.0284$	52	99.99	98.92
Case II	Ref. [39]	$C=1.8521$ $\sigma=0.7462$	–	96.54	99.99
	Ref. [21]	$C=0.9054$ $\sigma=0.7512$	–	95.65	100
	Ref. [40]	$C=14.3954$ $\sigma=1.9504$	173	98.75	96.58
	Recommended tactic	$C=16.0847$ $\sigma=1.2265$ $\xi=0.9574$	38	99.08	99.82

prediction compared to other methods, highlighting its stability in guessing the state of a data sample across changed executions. Another noteworthy point from Table 7 is the comparison of feature vectors obtained by the third method and the technique presented in this article. The third method selects a large number of features, leading to higher computational costs. In contrast, the feature selection approach recommended in this exploration results in a feature matrix with significantly smaller dimensions, reducing computational overhead. Specifically, for troubleshooting cases one to three, the features 128 F and 129 F were selected as optimal, whereas for the fourth case study, the features F127, F31, and 131 F were identified as optimal. This efficient feature selection not only improves the computational efficiency but also maintains high accuracy in fault detection. Overall, the recommended tactic demonstrates excellent performance regarding accuracy, stability, and computational efficiency, making it a robust choice for fault detection in bearings.

Conclusions

In this article, a new method for troubleshooting multi-class bearings with various defects at different rotational speeds is presented. The recommended tactic involves extracting vibration signals using EMD and WPD algorithms. The CDET is employed to select the optimal features.

Given that the CDET includes a crucial threshold parameter that determines the count of suitable features, this research uses the PSO algorithm to calculate this threshold value and identify the optimal feature set with the least number of features. This improved approach, referred to as the OCDET, enhances the feature selection process.

Subsequently, the most appropriate selected feature vector is used as input to an improved SVM algorithm. The new combined method is applied for fault diagnosis in four cases: fault type detection, inner rim fault, outer rim fault, and rolling element fault, each with different defect sizes.

The outcomes establish that the developed OCDET technique allows for the selection of the most suitable set of features, enabling better distinction between different bearing conditions. Utilizing the new feature set, the SVM accurately predicts the conditions of a bearing at various speeds. Furthermore, comparing the outcomes of this exploration with previous research validates the effectiveness and capability of the suggested technique.

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Authors' contributions
HC performed the data collection. XM carried out the simulation, and FZ conducted the analysis. YH evaluated the first draft of the manuscript, editing, and writing.

Data availability
Data are accessible upon request.

Declarations

Competing interests
The authors declare that they have no competing interests.

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